

Dylan Isaac

AI/LLM Systems Engineer • AI Product Engineer

github.com/dylan-isaac

Professional Summary

Engineer/designer building **agentic LLM systems** that turn messy, human workflows into **context-aware, explainable automation**. Ship features using **RAG**, **multi-agent voting**, and **human-in-the-loop** review to improve **accuracy** and reduce **false positives**. Accessibility is my proving ground; the patterns generalize to any domain where intent must be translated into **precise, adaptive behavior**.

Core Skills

- **LLM/AI Systems:** RAG pipelines; prompt/program design; **multi-agent voting**; human-in-the-loop review; intent→action translation; machine-readable knowledge corpora
- **Automation & Interop:** Agent workflows across teams/tools; role-aware guidance; cross-app **context sharing**
- **Modeling & Data:** **Graph-based** accessibility modeling; vectorized knowledge for retrieval/fine-tuning
- **Domain Expertise:** Accessibility testing/remediation; WCAG; assistive tech (VoiceOver, JAWS, NVDA)

Selected AI Projects

axe Assistant (Deque) — AI accessibility consultant for Slack/Teams

- Delivered **context-aware, role-specific remediation** using **RAG + expert heuristics**
- Introduced **multi-agent voting**, improving detection **~80%** and cutting **false positives to ~1%**

Deque Corpus Pipeline — Enterprise accessibility knowledge → vectorized, machine-readable formats

- Enabled precise AI recommendations and consistent guidance across roles

AI-Driven Guidance System — Intent-aware suggestions for designers, devs, and content teams

- Expanded AI-powered testing targeting historically hard-to-detect WCAG violations

Professional Experience

Deque Systems — Senior AI Accessibility Engineer & Architect

Remote | Aug 2023 – Present

- Invented & led development of **axe Assistant**, combining **RAG** and expert heuristics for role-specific remediation
- Architected the **Deque Corpus** to power retrieval and guidance with **vectorized knowledge**
- Pioneered **multi-agent voting**, boosting accuracy **~80%**, reducing FPs to **~1%**
- Launched initiatives for **AI-powered accessibility testing** on hard-to-detect issues
- Led internal hackathons/workshops; trained **100+** employees on leveraging AI

Deque Systems — Accessibility Coach

Remote | Aug 2022 – Aug 2023

- Coached **200+** at **PNC C&IB**; (design, content, dev) with role-specific workflows
- Built individualized learning paths, AT training, and advanced WCAG tracking
- Ran office hours/workshops that scaled adoption across multiple product lines

Everactive — Senior Software Engineer | Accessibility Lead

Charlottesville, VA | Feb 2021 – Jul 2022

- Collaborated with ML/hardware teams on accessible **IoT interfaces** with real-time data
- Trained in **vibration analysis / FFT** to refine **AI-driven UI alerts** for industrial monitoring
- Bridged hardware constraints, ML insights, and front-end UX for a unified, accessible experience

Coursera — Course Instructor

Remote | Jun 2020 – Aug 2020

- Built and taught an **Introduction to Accessible Web Development** course to **5,000+** learners
- Interactive modules on semantic HTML, WAI-ARIA, and automated testing; **4.8★** instructor rating

Skafos.ai — Software Engineer

Charlottesville, VA | Jun 2020 – Feb 2021

- Re-platformed front-end to **React micro-frontends**; built an accessibility-first component library
- Integrated **AI-powered product recommendations** with a11y-conscious UI

ChartIQ (interop.io) — Solutions Engineer

Charlottesville, VA | Sep 2019 – Apr 2020

- Built **contextual computing** workflows to synchronize user intent across apps
- Contributed to **FDC3** cross-application interoperability; applied those principles to AI workflows
- Delivered custom integrations for time-series analysis across finance/industry use cases

Early Career — Software Engineer & Accessibility Specialist

2016 – 2019 (Various)

- Legal, API, and university-library projects; standardized testing frameworks that inform today's **AI-assisted WCAG** strategies

Community & Standards

- **W3C Contributor**, **ARIA-AT User Group** • **Public speaking** on AI & accessibility

Tools & Technologies

AI: LLMs, RAG, Pydantic AI, MCP, multi-agent workflows, human-in-the-loop systems

Programming & Systems: Python, TypeScript, FastAPI, React, information architecture

Assistive Tech & Testing: VoiceOver, JAWS, NVDA, manual WCAG testing

Education

B.S., Information Science — Christopher Newport University

Selected Writing

[Epistemological Translator](#) • [Building Ladders](#)